

FACULTY OF COMPUTING & INFORMATICS (FCI)

CSN6224 COMPUTER NETWORKS

ASSIGNMENT

Trimester October/November 2025, Term 2530

GROUP 35

GROUP 35

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1.0 IP addressing plan

This table summarizes the IPv4 and IPv6 addressing scheme used in the network.

IPv4 addresses are assigned using /24 subnets for each network segment, while IPv6 addresses use /64 prefixes. Routers are configured with static addresses to act as default gateways, servers use static dual-stack addressing, and end devices obtain addresses dynamically via DHCP (IPv4) or SLAAC/DHCPv6 (IPv6).

Device	Interface	IPv4 Address/Prefix	IPv6 Address/Prefix	Default Gateway
Router 1	G0/0	202.10.4.254/24	2001:DB8:1003:2004::1/64	N/A
	G0/1.100	202.10.3.254/24	2001:DB8:1003:2005::1/64	N/A
	G0/2	202.18.11.1/24	2001:DB8:11::1/64	N/A
Router 2	G0/0.10	202.18.1.254/24	N/A	N/A
	G0/1	202.18.12.1/24	2001:DB8:12::1/64	N/A
	G0/2	202.18.11.2/24	2001:DB8:11::2/64	N/A
Router 3	G0/0.20	N/A	2001:DB8:1903:2004::1/64	N/A
	G0/1	202.18.12.2/24	2001:DB8:12::2/64	N/A
Router 4	G0/0	202.18.11.3/24	2001:DB8:11::3/64	N/A
	G0/1	172.16.1.1/24	2001:DB8:3::1/64	N/A
PC0	NIC	DHCP Assign	N/A	202.18.1.254
PC1	NIC	DHCP Assign	N/A	202.18.1.254
PC4	NIC	DHCP Assign	N/A	202.18.1.254
PC5	NIC	DHCP Assign	N/A	202.18.1.254
DHCP Server	NIC	202.18.1.100/24	N/A	202.18.1.254
PC31	NIC	N/A	SLAAC from 2001:DB8:1903:2004::/64	2001:DB8:1903:2004::1
PC32	NIC	N/A	SLAAC from 2001:DB8:1903:2004::/64	2001:DB8:1903:2004::1
DHCPv6	NIC	N/A	2001:DB8:1903:2004::10/64	2001:DB8:1903:2004::1
Web Server	NIC	202.10.4.10/24	2001:DB8:1003:2004::100/64	IPv4: 202.10.4.254 IPv6: 2001:DB8:1003:2004::1
DNS Server	NIC	202.10.3.11/24	2001:DB8:1003:2005::100/64	IPv4: 202.10.3.254 IPv6: 2001:DB8:1003:2005::1
Email Server	NIC	202.10.3.12/24	2001:DB8:1003:2005::101/64	IPv4: 202.10.3.254 IPv6: 2001:DB8:1003:2005::1
FTP Server	NIC	202.10.3.13/24	2001:DB8:1003:2005::102/64	IPv4: 202.10.3.254 IPv6: 2001:DB8:1003:2005::1
Radius & DHCP Server	NIC	172.16.1.10 /24	SLAAC from 2001:DB8:3::/64	IPv4: 172.16.1.1 IPv6: 2001:DB8:3::1
Laptop Marketing	Wireless	DHCP Assign	SLAAC from 2001:DB8:3::/64	IPv4: 172.16.1.1 IPv6: 2001:DB8:3::1
Laptop Engineering	Wireless	DHCP Assign	SLAAC from 2001:DB8:3::/64	IPv4: 172.16.1.1 IPv6: 2001:DB8:3::1
Laptop Guest	Wireless	DHCP Assign	SLAAC from 2001:DB8:3::/64	IPv4: 172.16.1.1 IPv6: 2001:DB8:3::1

2.0 Subnetting Tables

This table summarizes the IPv4 and IPv6 subnet allocation for all network segments, including user LANs, server networks, wireless networks, and core transit links. Each subnet is assigned a unique address range with clearly defined default gateways to support end-to-end connectivity across the network

Network Segment	IP Ver.	Network / Prefix	Mask / Prefix Length	Usable Host Range / Assignment	Broadcast (IPv4)	Default Gateway
User LAN IPv4 (VLAN 10)	IPv4	202.18.1.0/24	255.255.255.0	202.18.1.1 – 202.18.1.253	202.18.1.255	202.18.1.254
Server VLAN 100 (DNS/Email/FTP)	IPv4	202.10.3.0/24	255.255.255.0	202.10.3.1 – 202.10.3.253	202.10.3.255	202.10.3.254
Web LAN	IPv4	202.10.4.0/24	255.255.255.0	202.10.4.1 – 202.10.4.253	202.10.4.255	202.10.4.254
Wireless LAN	IPv4	172.16.1.0/24	255.255.255.0	172.16.1.2 – 172.16.1.254	172.16.1.255	172.16.1.1
Transit-1 (Core: R1–R2–R4)	IPv4	202.18.11.0/24	255.255.255.0	202.18.11.1 – 202.18.11.3	202.18.11.255	N/A
Transit-2 (R2–R3)	IPv4	202.18.12.0/24	255.255.255.0	202.18.12.1 – 202.18.12.2	202.18.12.255	N/A
User LAN IPv6 (VLAN 20)	IPv6	2001:DB8:1903:2004::/64	/64	SLAAC (RA)	N/A	2001:DB8:1903:2004::1
Wireless LAN (IPv6)	IPv6	2001:DB8:3::/64	/64	SLAAC (RA)	N/A	2001:DB8:3::1
Server LAN (IPv6)	IPv6	2001:DB8:1003:2005::/64	/64	Static (servers)	N/A	2001:DB8:1003:2005::1
Web LAN (IPv6)	IPv6	2001:DB8:1003:2004::/64	/64	Static (Web Server)	N/A	2001:DB8:1003:2004::1
Transit-1 (IPv6 Core)	IPv6	2001:DB8:11::/64	/64	Router interfaces	N/A	N/A
Transit-2 (IPv6 R2–R3)	IPv6	2001:DB8:12::/64	/64	Router interfaces	N/A	N/A

3.0 Configuration Steps with Screenshots

3.0.1 Router1: Dual-Stack Server Gateway & OSPF Configuration

This screenshot shows the complete configuration of Router1, including IPv4 and IPv6 addressing, VLAN 100 sub-interface setup, and OSPF/OSPFv3 routing.

Router1 is configured as the default gateway for the Web LAN and Server VLAN, and as a core router participating in dynamic routing to ensure full end-to-end connectivity across the network.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ipv6 unicast-routing
Router(config)#interface GigabitEthernet0/1
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1.100
Router(config-subif)#encapsulation dot1Q 100
Router(config-subif)#ip address 202.10.3.254 255.255.255.0
Router(config-subif)#ipv6 address 2001:DB8:1003:2005::1/64
Router(config-subif)#ipv6 ospf 1 area 0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 202.10.4.254 255.255.255.0
Router(config-if)#ipv6 address 2001:DB8:1003:2004::1/64
Router(config-if)#ip ospf 1 area 0
Router(config-if)#ipv6 ospf 1 area 0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#ip address 202.18.11.1 255.255.255.0
Router(config-if)#ipv6 address 2001:DB8:11::1/64
Router(config-if)#ip ospf 1 area 0
Router(config-if)#ipv6 ospf 1 area 0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 202.10.3.0 0.0.0.255 area 0
Router(config-router)#network 202.10.4.0 0.0.0.255 area 0
Router(config-router)#network 202.18.11.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#
Router(config)#ipv6 router ospf 1
Router(config-rtr)#router-id 1.1.1.1
Reload or use "clear ipv6 ospf process" command, for this to take effect
Router(config-rtr)#exit
Router(config)#
```

3.0.2 Router2: User LAN IPv4 (VLAN 10) Gateway & Core OSPF Configuration

This screenshot shows the complete configuration of Router2, including the IPv4 User LAN gateway for VLAN 10, the core transit interface, and OSPF/OSPFv3 routing.

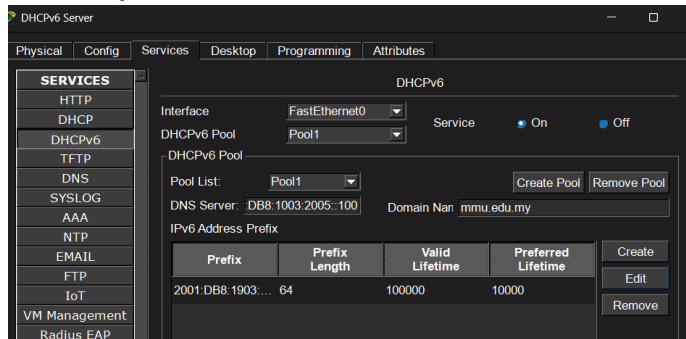
Router2 functions as the default gateway for the User LAN and participates in dynamic routing to exchange IPv4 and IPv6 routes with the core network, ensuring end-to-end connectivity.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface GigabitEthernet0/0
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface GigabitEthernet0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 202.18.1.254 255.255.255.0
Router(config-subif)# ip ospf 1 area 0
Router(config-subif)# no shutdown
Router(config-subif)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)# ip address 202.18.11.2 255.255.255.0
Router(config-if)# ipv6 address 2001:DB8:11::2/64
Router(config-if)# ip ospf 1 area 0
Router(config-if)# ipv6 ospf 1 area 0
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)# router-id 2.2.2.2
Router(config-router)# network 202.18.1.0 0.0.0.255 area 0
Router(config-router)# network 202.18.11.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#
Router(config)#ipv6 router ospf 1
Router(config-rtr)# router-id 2.2.2.2
Reload or use "clear ipv6 ospf process" command, for this to take effect
Router(config-rtr)#exit
Router(config)#
```

3.0.3 Router3: User LAN IPv6 (VLAN 20), DHCPv6 & OSPF Configuration

Router3 is configured as the IPv6 default gateway for the IPv6 User LAN (VLAN 20) and participates in OSPFv3 routing to enable IPv6 connectivity across the core network.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#ipv6 unicast-routing
Router(config)#
Router(config)#interface GigabitEthernet0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ipv6 address 2001:DB8:1903:2004::1/64
Router(config-subif)# ipv6 enable
Router(config-subif)# ipv6 ospf 1 area 0
Router(config-subif)# exit
Router(config)#
Router(config)#interface GigabitEthernet0/1
Router(config-if)# ipv6 address 2001:DB8:12::2/64
Router(config-if)# ipv6 enable
Router(config-if)# ipv6 ospf 1 area 0
Router(config-if)# exit
Router(config)#
Router(config)#ipv6 router ospf 1
Router(config-rtr)# router-id 3.3.3.3
Reload or use "clear ipv6 ospf process" command, for this to take effect
Router(config-rtr)# exit
Router(config)#
Router(config)#end
```



3.0.4 Router4: Wireless Gateway, Radius AAA & Console, VTY, and enable-secret security Configuration

This screenshot shows the complete configuration of Router4, which functions as the wireless gateway and a core network router. Router4 connects the wireless LAN to the core network using dual-stack IPv4 and IPv6 addressing and participates in dynamic routing through OSPF and OSPFv3. The configuration enables wireless clients to obtain network connectivity and communicate with other network segments through the core. Router4 is configured with AAA to integrate with the RADIUS server for authenticating wireless users.

```
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#aaa new-model
R4(config)#aaa authentication login TELNET_AUTH group radius local
R4(config)#aaa authentication login WLAN_AUTH group radius local
R4(config)#radius-server host 172.16.1.10 key cisco
R4(config)#line vty 0 4
R4(config-line)#login authentication TELNET_AUTH
R4(config-line)#transport input telnet
R4(config)#enable secret class123
R4(config)#service password-encryption
```

```

User Access Verification

Username: ml
Password:
R4>enable
Username:
Password:
R4#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#interface GigabitEthernet0/0
R4(config-if)# ip address 202.18.11.3 255.255.255.0
R4(config-if)# ipv6 address 2001:DB8:11::3/64
R4(config-if)#ip ospf 1 area 0
R4(config-if)# ipv6 ospf 1 area 0
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#interface GigabitEthernet0/1
R4(config-if)# ip address 172.16.1.1 255.255.255.0
R4(config-if)# ipv6 address 2001:DB8:3::1/64
R4(config-if)# no shutdown
R4(config-if)#exit
R4(config)#router ospf 1
R4(config-router)# router-id 4.4.4.4
R4(config-router)# network 202.18.11.0 0.0.0.255 area 0
R4(config-router)#exit
R4(config)#
R4(config)#ipv6 router ospf 1
R4(config-rtr)# router-id 4.4.4.4
Reload or use "clear ipv6 ospf process" command, for this to take
effect
R4(config-rtr)#exit

```

3.0.5 VLAN 10 Configuration – IPv4 User LAN (Switch 22)

VLAN 10 is used to separate IPv4 user traffic from other network segments.

User PCs connected to FastEthernet0/1 and FastEthernet0/2 are assigned to VLAN 10.

```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name VLAN0010
Switch(config-vlan)#exit
Switch(config)#interface fastEthernet0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit

```

3.0.6 VLAN 100 Configuration – Server LAN (Switch 11)

VLAN 100 isolates server traffic from user networks to improve security and performance.

Servers are connected to access ports configured under VLAN 100.

```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 100
Switch(config-vlan)#name VLAN0100
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#interface fastEthernet0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 100
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fastEthernet0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 100
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fastEthernet0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 100
Switch(config-if)#exit

```

3.0.7 VLAN 20 Configuration - IPv6 User LAN (Switch 31)

VLAN 20 is configured to isolate IPv6 user traffic from other network segments. IPv6 user devices are connected to access ports assigned to VLAN 20, and VLAN 20 traffic is forwarded to Router3 through a trunk link to enable IPv6 routing and SLAAC address assignment.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 20
Switch(config-vlan)# name IPv6_User_LAN
Switch(config-vlan)# exit
Switch(config)#
Switch(config)#interface fastEthernet0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
Switch(config)#
Switch(config)#interface fastEthernet0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
Switch(config)#
Switch(config)#interface fastEthernet0/3
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
Switch(config)#
```

3.0.8 Switch Configuration (VLAN & Trunk)

Trunk ports were configured with IEEE 802.1Q and verified using `show interfaces trunk`.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fastEthernet0/3
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
```

Trunk Verification

```
Switch>enable
Switch#show interfaces trunk

Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     desirable n-802.1q       trunking      1
Fa0/2     on        802.1q         trunking      1
Fa0/3     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/2     1-1005
Fa0/3     1-1005
```

3.0.9 RADIUS Server Configuration

The RADIUS server is configured to authenticate wireless users and store user credentials, with Router4 registered as a RADIUS client.

Radius & DHCP Server

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

AAA

Service ☒ On ☐ Off Radius Port 1645

Network Configuration

Client Name Client IP

Secret Server Type

	Client Name	Client IP	Server Type	Key	
1	R4	172.16.1.1	Radius	cisco	Add
					Save
					Remove

User Setup

Username Password

	Username	Password	
1	e1	2345	Add
2	guest	g1	
3	m1	1234	Save
			Remove

Top

3.0.10 DHCP Server Configuration

The DHCP server is enabled and configured to allocate IPv4 addresses from the 172.16.1.0/24 pool.

Radius & DHCP Server

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface Service ☒ On ☐ Off

Pool Name

Default Gateway

DNS Server

Start IP Address:

Subnet Mask:

Maximum Number of Users:

TFTP Server:

WLC Address:

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	172.16.1.1	202.10.3.11	172.16.1.100	255.255.255.0	156	0.0.0.0	0.0.0.0

4.0 Test results (IPv4 & IPv6)

IPv4 connectivity test from PC to Web Server

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 202.10.4.10

Pinging 202.10.4.10 with 32 bytes of data:

Reply from 202.10.4.10: bytes=32 time<1ms TTL=126
Reply from 202.10.4.10: bytes=32 time=13ms TTL=126
Reply from 202.10.4.10: bytes=32 time<1ms TTL=126
Reply from 202.10.4.10: bytes=32 time<1ms TTL=126

Ping statistics for 202.10.4.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 3ms
```

IPv4 connectivity test from PC to Wireless Laptop

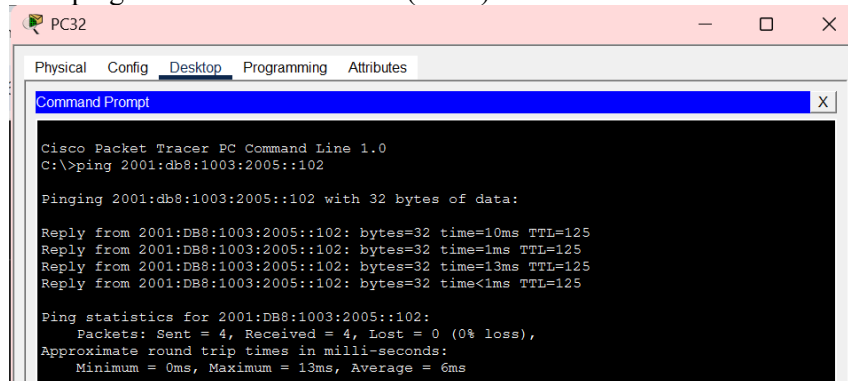

```
C:\>ping 172.16.1.102

Pinging 172.16.1.102 with 32 bytes of data:

Request timed out.
Reply from 172.16.1.102: bytes=32 time=36ms TTL=126
Reply from 172.16.1.102: bytes=32 time=10ms TTL=126
Reply from 172.16.1.102: bytes=32 time=80ms TTL=126

Ping statistics for 172.16.1.102:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 80ms, Average = 42ms
```

IPv6 ping test from IPv6 user PC (PC32) to the FTP server



IPv6 ping test from IPv6 user PC (PC32) to the wireless laptop

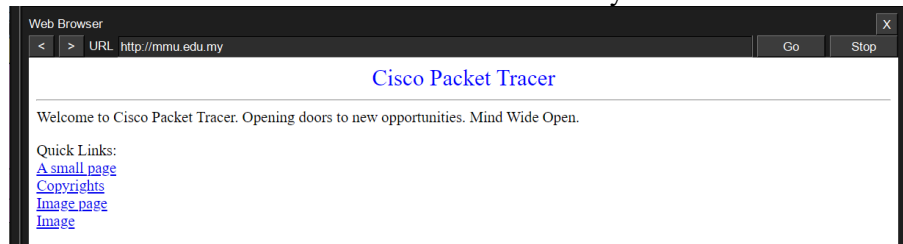
```
C:\>ping 2001:db8:3:0:202:4aff:fe65:c169

Pinging 2001:db8:3:0:202:4aff:fe65:c169 with 32 bytes of data:

Reply from 2001:DB8:3:0:202:4AFF:FE65:C169: bytes=32 time=52ms
TTL=125
Reply from 2001:DB8:3:0:202:4AFF:FE65:C169: bytes=32 time=19ms
TTL=125
Reply from 2001:DB8:3:0:202:4AFF:FE65:C169: bytes=32 time=10ms
TTL=125
Reply from 2001:DB8:3:0:202:4AFF:FE65:C169: bytes=32 time=10ms
TTL=125

Ping statistics for 2001:DB8:3:0:202:4AFF:FE65:C169:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 52ms, Average = 22ms
```

Successful HTTP access confirms IPv4 connectivity to the web server.

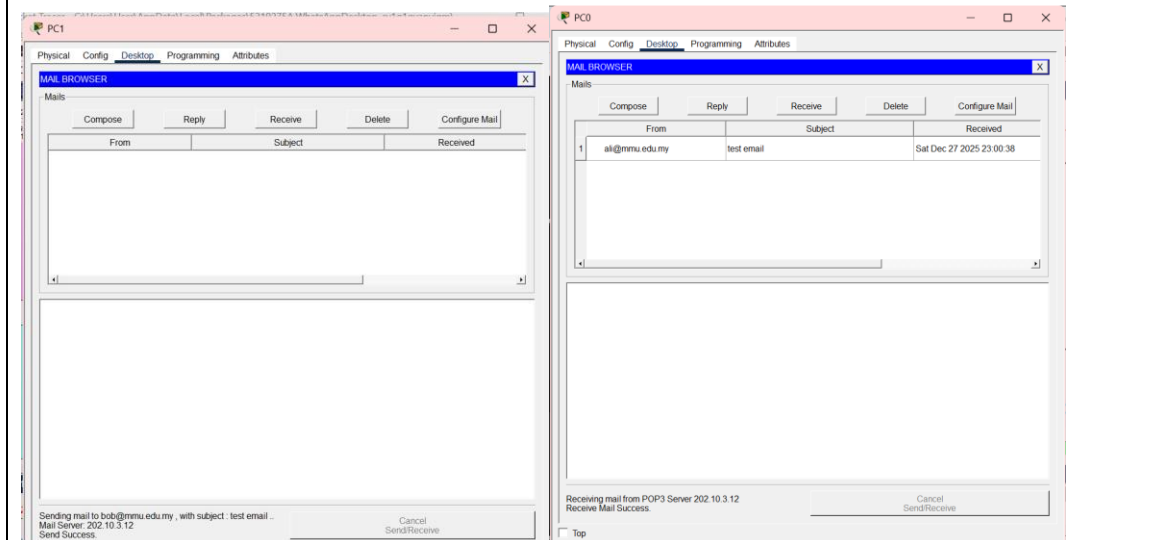


FTP login

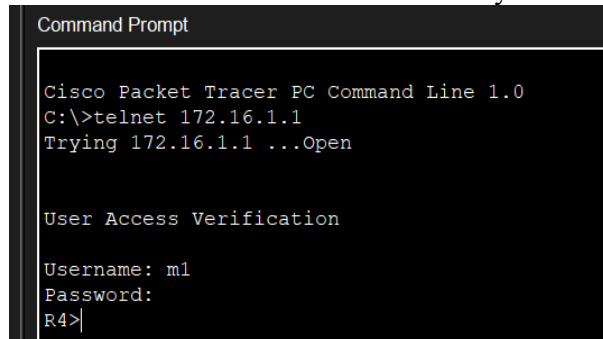
```
C:\>ftp 202.10.3.13
Trying to connect...202.10.3.13
Connected to 202.10.3.13
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 202.10.3.13:
 0 : asa842-k8.bin                      5571584
 1 : asa923-k8.bin                      30468096
 2 : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
 3 : c1841-ibase-mz.123-14.T7.bin       13832032
 4 : c1841-ibasek9-mz.124-12.bin        16599160
 5 : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
 6 : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
 7 : c2600-i-mz.122-28.bin              5571584
 8 : c2600-ibasek9-mz.124-8.bin          13169700
 9 : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
10 : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
11 : c2800nm-ibase-mz.123-14.T7.bin      5571584
12 : c2800nm-ibasek9-mz.124-8.bin        15522644
```

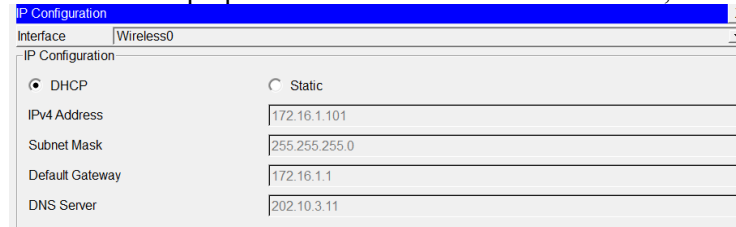
Email send and receive test successful.



RADIUS authentication was confirmed by a successful Telnet login to Router 4.



The wireless laptop receives an IPv4 address via DHCP, indicating successful network access.



5.0 Reflection and Lessons Learned

This project provided practical experience in designing and implementing a multi-network topology using both IPv4 and IPv6. Through the configuration of VLANs, routing protocols, and dual-stack addressing, a deeper understanding of network segmentation and inter-network communication was gained.

The integration of wireless networks with RADIUS authentication improved understanding of secure network access control and centralized user authentication. Configuring OSPF and OSPFv3 also highlighted the importance of dynamic routing for scalability and efficient network management.

Overall, this project strengthened troubleshooting skills, improved familiarity with Packet Tracer, and reinforced the importance of careful planning and documentation when designing complex networks.

Presentation: <https://youtu.be/gH8nKZVSE48>

Group Marking Rubric (90%)

Assignment

CSN6224 Communication Networks

Trimester October/November 2025 Term 2530

Group
Name:

Group 35

Group ID:

Group 35

This document contains the official assessment rubric for your assignment. Marks will be awarded based on the criteria and weightage specified in this rubric, which accounts for **90%** of the **total project marks**. All groups must follow this rubric when completing and presenting their work. The remaining **10%** of the marks will be based on the **Individual Self-Evaluation Form** submitted together with your final report.

Criteria	Weightage (%)	Score 4	Score 3	Score 2	Score 1	Score 0	Rating (Lecturer)
1. VLAN Segmentation & Switch Configuration	15%	VLANs, trunks, STP, redundancy fully correct & stable	Mostly correct with minor errors	VLANs correct but STP/trunks incomplete	Major errors in VLAN or trunk config	Not attempted	
2. IPv4 & IPv6 Addressing Plan (DOB-Based)	10%	Fully correct IPv4/IPv6 addressing with accurate tables	Mostly correct	Some errors or incomplete tables	Incorrect formatting or wrong addressing	Not attempted	
3. Dual-Stack Server Configuration	10%	All services (DNS, Web, Email, FTP) fully functional on IPv4 & IPv6	Most services work; minor issues	Some services functional	Only 1 service working / major issues	No servers configured	
4. Dynamic Routing (IPv4 + IPv6)	10%	Full OSPFv2/OSPFv3 (or RIP) convergence across all routers	Routing works but minor gaps	Partial routing; some unreachable networks	Incorrect routing protocols or major errors	No dynamic routing	
5. Wireless + RADIUS Authentication	10%	All 3 WLANs functional with stable WPA2-Enterprise, DHCP & RADIUS	RADIUS mostly works with minor issues	WLAN OK, RADIUS partially working	Incorrect WLAN/RADIUS setup	Not attempted	
6. End-to-End Connectivity & Simulation Testing	10%	All IPv4/IPv6 ping, services, RADIUS login, simulation verified	Most tests successful	Partial connectivity	Few tests working	No testing conducted	
7. Report Quality & Documentation	10%	Professional, clear, complete, with appendices & evidence	Well-written and complete	Adequate but lacks depth or clarity	Poorly organized	No report submitted	
8. Presentation (Slides + Video)	10%	Excellent clarity, structure, teamwork, technical accuracy	Good delivery with minor gaps	Basic presentation	Poor clarity or missing parts	No presentation submitted	
9. Bonus: Enhancements (SSH, ACLs, Redundancy, Syslog/SNMP)	5%	3+ enhancements fully functional	Multiple enhancements with minor issues	1-2 enhancements partially working	Attempted but incorrect	No enhancements	

Weekly Project Guidelines / Checklist (Week 5 – 13)

Assignment

CSN6224 Communication Networks
Trimester October/November 2025 Term 2530

Group
Name:

Group 35

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These guidelines serve as a weekly reminder of the suggested tasks for your group. They are intended to help you plan and complete your project effectively. You are not required to follow them strictly, and your group may adjust the schedule as needed based on progress and workload.

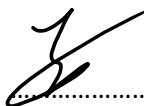
Week	Task Title	Task Item	Checklist
Week 5	Group Formation	Form group & confirm member list	Checked
		Set up communication channel	Checked
		Assign roles (Leader, Technical Lead, Documentation Lead, Presentation Lead, QA)	Checked
		Conduct first meeting	Checked
Week 6	Requirements Analysis	Study assignment requirements	Checked
		Develop project scenario	Checked
		Divide tasks among group members	Checked
		Prepare high-level network design	Checked
		Lecturer check-in: confirm scope and design idea	-
Week 7	Initial Network Design	Draft topology in Packet Tracer	Checked
		Start IP addressing plan	Checked
		Identify required protocols	Checked
		Lecturer check-in: verify topology and addressing draft	-
Week 8	Requirements Mapping	Finalize topology	Checked
		Begin basic configurations	Checked
		Lecturer check-in: confirm all requirements covered	-

Week	Task Title	Task Item	Checklist
Week 9	Core Implementation	Configure routing, switching, DHCP/NAT/ACL	Checked
		Conduct basic connectivity tests	Checked
		Lecturer check-in: demonstrate partial working network	-
Week 10	Troubleshooting & Enhancement	Fix issues and refine configurations	Checked
		Add enhancements (if allowed)	2 added
		Lecturer check-in: verify functionality	-
Week 11	Documentation & Review	Finalize topology, tables, configuration summaries, test results	Checked
		QA check	Checked
		Lecturer check-in: review near-final work	-
Week 12	Final Report & Video	Finalize written report	Checked
		Finalize presentation slides	Checked
		Record project video presentation	Checked
		Upload video to YouTube (Unlisted)	Checked
Week 13	Submission to eBwise	Submit Packet Tracer file (.pkt)	Checked
		Submit PDF report	Checked
		Submit presentation slides (.pptx or other format)	Checked

Group Name: Group 35

NO.	TASK DESCRIPTION	MEMBER-IN-CHARGE	DATE ASSIGNED	DATE COMPLETED	REMARK
1	IPv4 User Network	EE YU SUEN	1/5/2026	20/1/2026	
2	IPv6 User Network	EE YU SUEN	1/5/2026	20/1/2026	
3	High Availability Core Network	EE YU SUEN	1/5/2026	20/1/2026	
4	Dual Stack Network	EE YU SUEN	1/5/2026	20/1/2026	
5	Additional Network – Wireless + RADIUS	EE YU SUEN	1/5/2026	20/1/2026	
6	IPv4 & IPv6 Addressing	EE YU SUEN	1/5/2026	20/1/2026	
7	VLAN	EE YU SUEN	1/5/2026	20/1/2026	
8	Console, VTY, and enable-secret security	EE YU SUEN	1/5/2026	26/1/2026	
9	Router Configuration	BEATRICE LEW YU LING	1/5/2026	22/1/2026	
10	Testing & Validation	TAN YI LING	1/5/2026	27/1/2026	
11	Report	TAN YI LING	1/5/2026	27/1/2026	
12	PPT	BEATRICE LEW YU LING	1/5/2026	28/1/2026	
13	Video Presentation	BEATRICE LEW YU LING	1/5/2026	28/1/2026	

Prepared by:



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Group Leader

Name: Beatrice Lew Yu Ling

ID : 242UC244R9

Individual Self-Evaluation Form (10%)

Assignment

CSN6224 Communication Networks
Trimester October/November 2025 Term 2530

Student Name:	Beatrice Lew Yu Ling	Student ID:	242UC244R9
Group Name:	Group 35	Group ID:	Group 35

Each group member must complete and sign this form to verify the accuracy of their self-evaluation.

Please **attach** the completed form together with your group report during submission.

If you have **concerns** or **disagree** with any part of a group member's self-evaluation, please **email** the lecturer **privately**. The lecturer will review the matter, and appropriate action will be taken if the concern is substantiated.

Criteria	Weightage (%)	4 – Excellent	3 – Good	2 – Fair	1 – Poor	0 – None	Your Rating (Number)
Attendance	2	Attended all meetings except excused official tasks (with proof).	Missed 1 meeting with acceptable reason.	Missed 2–3 meetings; explanation acceptable but weak.	Frequently absent; weak or unclear reasons.	No attendance / no explanation.	4
Task Completion	3	Completed all tasks on time with clear evidence.	Completed most tasks; minor delays.	Completed some tasks; multiple delays.	Very minimal task completion.	No completed tasks.	4
Communication	1.5	Always responsive; active in chat & group activities.	Mostly responsive; occasional delay.	Sometimes responds; inconsistent.	Rarely responds; slow communication.	No communication at all.	4
Contribution	2	Actively contributes ideas and solutions in discussions.	Regular contributions; useful input.	Occasional contributions; basic input.	Rarely contributes; passive.	No contribution.	4
Initiative	1.5	Proactively initiates tasks and helps others consistently.	Often shows initiative and helps when needed.	Sometimes helps; limited initiative.	Rarely initiates; only helps when asked.	No initiative or avoids helping.	4

By signing below, you confirm that all information provided in this form is true and accurate.

Declaration



I hereby confirm that the above self-evaluation is accurate.



Student Name: Beatrice Lew Yu Ling

Student ID: 242UC244R9

Date: 29/1/2026

Individual Self-Evaluation Form (10%) Assignment

**CSN6224 Communication Networks
Trimester October/November 2025 Term 2530**

Student Name:	Ee Yu Suen	Student ID:	242UC244FH
Group Name:	Group 35	Group ID:	Group 35

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Declaration



I hereby confirm that the above self-evaluation is accurate.



Student Name: Ee Yu Suen

Student ID: 242UC244FH

Date: 29/1/2026

Individual Self-Evaluation Form (10%)

Assignment

CSN6224 Communication Networks

Trimester October/November 2025 Term 2530

Student Name:	Tan Yi Ling
Group Name:	Group 35

Student ID:	242UC244PD
Group ID:	Group 35

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Student Name: Tan Yi Ling

Student ID: 242UC244PD

Date: 29/1/2026